

Building a CNN for Rock-Paper-Scissors Classification

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Abstract. This project aims to build and compare three different convolutional neural networks (CNNs) for classifying images of hand gestures representing rock, paper, scissors. The three CNNs represent three increasing levels of complexity and were trained on the same standardized dataset containing 2188 images of hand gestures. The training was performed using a train test split of 70/30. Testing was performed using both the test set from the standardized dataset and two custom datasets containing 70 and 40 images respectively, which were collected by the author and contain more challenging images (e.g. with different backgrounds, lighting conditions, hand orientations, etc.). Results show that the most complete CNN achieved the best performance on the standardized test set, with higher validation accuracy than training accuracy, which suggests that the model is not overfitting and is able to generalize well to unseen data. However, the performance on the custom datasets was significantly lower, especially on the second custom dataset which contains images with different backgrounds where the model's performance is similar to a random classifier. Analyzing the results, it can be observed that the model is able to correctly classify images with simple backgrounds and good lighting conditions, but struggles with images that have more complex backgrounds or poor lighting conditions. Another critical aspect is the position of the fingers, error analysis revealed that most of the misclassifications contain finger positions that are similar to other classes.

1 Dataset Exploration

The main dataset used for this project is the Kaggle Rock-Paper-Scissors dataset [6], which contains 2,188 images of hand gestures representing rock, paper, and scissors. The dataset is organized into three folders, one for each class, and each folder contains images with similar orientations and lighting conditions. The images are provided in JPEG format with a resolution of 300×200 pixels.

In addition to the main dataset, two secondary custom datasets were collected by the author, containing 40 and 70 images respectively. These datasets were used to evaluate the performance of the CNNs on more challenging and diverse images. The first custom dataset includes hand gesture images captured under different lighting conditions and with varying rotations, while still maintaining a homogeneous background. The second custom dataset contains images with

more heterogeneous backgrounds, different camera angles, and greater variability overall. Both custom datasets are balanced, with approximately the same number of images for each class.

All datasets contain the same three classes: rock, paper, and scissors. However, the primary training dataset is highly homogeneous, as most images share similar orientations, poses, lighting conditions, and backgrounds. This homogeneity is reflected in the high performance achieved by the model on the training data. The same limitation also affects the validation and test sets, since they are generated by splitting the same original dataset used for training. As a result, these sets may not fully represent the variability encountered in real-world scenarios, making generalization to more diverse images more challenging.



Fig. 1. Sample images from the training dataset.

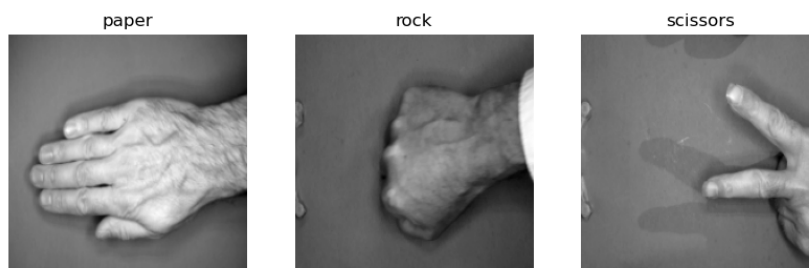
2 Methodology

2.1 Preprocessing

To speed up both processing and testing the images in the dataset were resized to 200x200 pixel and changed to greyscale, this allows to process only one third of the data for each image, since the color channels are reduced to one. This also means that the model will not learn any color-based feature, in this particular usecase this is a good thing since the classification is strictly shape based, color could not influence the decisions of the model and could even be a source of overfitting, since the training dataset is very homogeneous and the images all share the same green background.

Images were then normalized by dividing their pixel values by 255 to scale them to the range $[0, 1]$. This normalization step helps to improve the convergence of the CNN during training by ensuring that the input data is on a consistent scale. This step was implemented in the code as a layer within the model, but conceptually it is a preprocessing step that is applied to the input data before it is fed into the CNN.

The main dataset was split into a training set (70%), a validation set (15%), and a test set (15%). The training set was used to train the CNNs, while the validation set was used to tune hyperparameters and prevent overfitting. The test set was reserved for evaluating the final performance of the models after training.



2.2 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

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Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

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